Differential Equations and Slope Fields

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Verifying solutions to differential equations**

Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C.

$\frac{dy}{dx} = 3Ce^{3x} = 3(Ce^{3x}) = 3y$. This confirms $y = Ce^{3x}$ satisfies the equation for any C.

2 **[Foundational]** • **Target: Verifying solutions to differential equations**

Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.

$\frac{dy}{dx} = 2x$ ✓. Using $y(1) = 5$: $1 + C = 5 \Rightarrow C = 4$. Particular solution: $y = x^2 + 4$.

3 **[Foundational]** • **Target: Solving separable differential equations**

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

$y\,dy = x\,dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 3$: $9 = 0 + 2C \Rightarrow 2C = 9$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (positive root, since $y(0) = 3 > 0$).

4 **[Foundational]** • **Target: Slope fields**

For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, as would be drawn on a slope field.

At $(0, 0)$: slope $= 0 - 0 = 0$. At $(1, 0)$: slope $= 1 - 0 = 1$. At $(1, 1)$: slope $= 1 - 1 = 0$.

5 **[Foundational]** • **Target: Slope fields**

Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.

For $k > 0$: slopes are positive above the x-axis and negative below, steepening as $|y|$ increases — curves grow away from $y = 0$ (unstable equilibrium, exponential growth). For $k < 0$: slopes point toward $y = 0$ from both sides, flattening as $|y| \to 0$ — curves decay toward $y = 0$ (stable equilibrium, exponential decay).

6 **[Foundational]** • **Target: Exponential growth and decay models**

A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.

Solution form: $P(t) = P_0 e^{kt} = 500e^{0.2t}$. $P(10) = 500e^2 \approx 500(7.389) \approx 3694$ bacteria.

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7 [Foundational] • Target: Exponential growth and decay models

A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).

$A(t) = 100e^{-0.05t}$. Set $50 = 100e^{-0.05t}$: $0.5 = e^{-0.05t}$, $\ln(0.5) = -0.05t$, $t = \frac{\ln(0.5)}{-0.05} = \frac{-0.693}{-0.05} \approx 13.86$ years.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Matching slope fields to differential equations

A slope field shows horizontal segments (slope 0) along the entire x-axis and along the entire y-axis, with slopes elsewhere resembling the product xy. Identify the differential equation this slope field represents.

$\frac{dy}{dx} = xy$ — slope is 0 whenever $x = 0$ or $y = 0$, matching the described field.

9 [Intermediate] • Target: Newton's Law of Cooling

An object cools according to $\frac{dT}{dt} = -k(T - 70)$, where $70°$F is room temperature. If $T(0) = 200°$F and $k = 0.1$, find $T(t)$.

Let $y = T - 70$: $\frac{dy}{dt} = -0.1y \Rightarrow y = y_0e^{-0.1t}$. $y_0 = 200 - 70 = 130$. So $T(t) = 70 + 130e^{-0.1t}$.

10 [Intermediate] • Target: Visualizing an exponential growth solution

The graph shows the bacteria population $P(t) = 500e^{0.2t}$ from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using $\ln 2$.

The graph crosses $P = 1000$ at approximately $t \approx 3.5$. Algebraically: $1000 = 500e^{0.2t}$ gives $2 = e^{0.2t}$, so $t = \frac{\ln 2}{0.2} \approx 3.47$, matching the graphical estimate.

11 [Intermediate] • Target: A logistic-style differential equation

A population satisfies $\frac{dP}{dt} = 0.1P\left(1 - \frac{P}{200}\right)$. Find the equilibrium solutions (where $\frac{dP}{dt} = 0$), and explain which one is stable for a population starting near $P = 10$.

Setting $\frac{dP}{dt} = 0$: $P = 0$ or $P = 200$. Starting near $P = 10$ (close to 0 but positive), the population will grow toward the stable equilibrium $P = 200$ (the carrying capacity), since $P = 0$ is unstable for any positive starting population.

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12 **[Intermediate]** • **Target: A mixing problem modeled by a differential equation**

A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If $y(t)$ is the amount of salt (kg) at time t, the differential equation is $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$. Find the equilibrium value of y, and explain what it represents physically.

Setting $\frac{dy}{dt} = 0$: $6 = \frac{y}{25}$ gives $y = 150$ kg. This represents the long-run salt content: eventually the rate of salt entering (6 kg/min from the inflow) balances the rate leaving, keeping the amount constant at 150 kg.

13 **[Intermediate]** • **Target: Slope fields**

Level 1 · Foundations Find the particular solution to $\frac{dy}{dx} = \frac{3x^2}{y}$ subject to $y(0) = 4$.

Separate variables: $y\,dy = 3x^2\,dx$. Integrate both sides: $\int y\,dy = \int 3x^2\,dx \Rightarrow \frac{1}{2}y^2 = x^3 + C$. Apply initial condition $y(0) = 4$: $\frac{1}{2}(4)^2 = 0 + C \Rightarrow C = 8$. Multiply by 2: $y^2 = 2x^3 + 16$. Since $y(0) = 4 > 0$, choose the positive root: $y = \sqrt{2x^3 + 16}$.

14 **[Intermediate]** • **Target: Exponential growth and decay models**

Level 1 · FoundationsA population satisfies $\frac{dP}{dt} = 0.05P - 0.0001P^2$. Determine the carrying capacity L and evaluate $\lim_{t\to\infty} P(t)$ given initial condition $P(0) = 150$.

Factor the differential equation: $\frac{dP}{dt} = 0.0001P(500 - P)$. The carrying capacity is $L = 500$. Since $P_0 = 150 > 0$, as $t \to \infty$, the population approaches its carrying capacity asymptotically: $\lim_{t\to\infty} P(t) = \mathbf{500}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Matching slope fields to differential equations** • **Cross-Lesson: 8.01 Area Between Curves**

[Differential Equations · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

16 **[Challenging]** • **Target: Newton's Law of Cooling** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

[Unit 7BC · Logistic Models & Euler's Method · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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17 **[Challenging]** • **Target: Visualizing an exponential growth solution** • **Cross-Lesson: 1.01 Limits Graphically and Numerically**

A weighted network has edges: DE = 1, BC = 2, AB = 3, CD = 4, BD = 5, CE = 6, AC = 7, AD = 8. Use Kruskal's algorithm to determine the minimum spanning tree and compute its total weight.

1. Sort all edges in non-decreasing order of weight: DE (1), BC (2), AB (3), CD (4), BD (5), CE (6), AC (7), AD (8). 2. Select edges avoiding cycles: - Add DE (weight 1) [Vertices connected: {D, E}] - Add BC (weight 2) [Vertices connected: {D, E}, {B, C}] - Add AB (weight 3) [Vertices connected: {D, E}, {A, B, C}] - Add CD (weight 4) [Merges components: {A, B, C, D, E} - all 5 vertices connected!] 3. Check edge count: $V - 1 = 5 - 1 = 4$ edges selected. 4. Total MST Weight: $1 + 2 + 3 + 4 = 10$.

18 **[Challenging]** • **Target: A logistic-style differential equation** • **Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

Level 2 · MediumFind the slope of the tangent line $\frac{dy}{dx}$ to the polar cardioid $r = 1 - \cos\theta$ at $\theta = \frac{\pi}{2}$.

Recall $x = r\cos\theta$ and $y = r\sin\theta$. By the Product Rule: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$. Given $r = 1 - \cos\theta \Rightarrow r' = \sin\theta$. At $\theta = \pi/2$: $r = 1 - 0 = 1$, $r' = 1$, $\sin(\pi/2) = 1$, $\cos(\pi/2) = 0$. Substitute: $\frac{dy}{dx} = \frac{(1)(1) + (1)(0)}{(1)(0) - (1)(1)} = \frac{1}{-1} = \mathbf{-1}$.

19 **[Challenging]** • **Target: Additional separable equations** • **Cross-Lesson: 2.01 The Definition of the Derivative**

A network has weighted adjacency matrix W. Explain how the minimum spanning tree algorithm (Kruskal's) differs from Dijkstra's algorithm.

Kruskal's algorithm finds a global minimum spanning tree that connects all vertices in the graph with minimum total edge weight, without reference to any single starting vertex. Dijkstra's algorithm finds the shortest path routes from a designated single source vertex to all other individual vertices. A minimum spanning tree does not necessarily contain the shortest paths between arbitrary pairs of vertices.

20 **[Challenging]** • **Target: A mixing problem modeled by a differential equation** • **Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives**

Level 1 · FoundationsConvert the Cartesian equation $x^2 + y^2 = 6y$ into polar form, and identify the resulting geometric shape.

Substitute $x^2 + y^2 = r^2$ and $y = r\sin\theta$: $r^2 = 6r\sin\theta \Rightarrow \mathbf{r = 6\sin\theta}$. In Cartesian coordinates, $x^2 + (y^2 - 6y + 9) = 9 \Rightarrow x^2 + (y - 3)^2 = 3^2$, which is a circle with radius 3 centered at $(0, 3)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Verifying solutions to differential equations

For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(-1, -4)$, as would be drawn on a slope field.

a. -19

b. -17 ✓

c. -15

d. 23

Slope $= -3(-1) - 20 = 3 - 20 = -17$.

22 [Foundational] • Target: Solving separable differential equations

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 21$. Find $y(20)$ (the positive root).

a. 21

b. 29 ✓

c. 20

d. 28

$y\,dy = x\,dx \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 21$: $2C = 441$. So $y^2 = x^2 + 441$;
$y(20) = \sqrt{(20)^2 + 441} = \sqrt{841} = 29$.

23 [Foundational] • Target: Slope fields

A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 326$. Find $P(5)$, rounded to the nearest whole number.

a. 587

b. 594

c. 794

d. 690 ✓

$P(t) = P_0 e^{kt} = 326e^{\frac{3}{20}t}$. $P(5) = 326e^{\frac{3}{20}(5)} \approx 690$.

24 [Foundational] • Target: Exponential growth and decay models

A quantity decays according to $\frac{dP}{dt} = -\frac{1}{4}P$, with $P(0) = 473$. Find $P(7)$, rounded to the nearest whole number.

a. 106

b. 95

c. 70

d. 82 ✓

$P(t) = P_0 e^{kt} = 473e^{-\frac{1}{4}t}$. $P(7) = 473e^{-\frac{1}{4}(7)} \approx 82$.

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25 [Foundational] • Target: Matching slope fields to differential equations

A substance decays according to $\frac{dA}{dt} = -\frac{1}{10}A$. If $A(0) = 126$ grams, find how long it takes to decay to 84 grams, to one decimal place.

a. 9.1

b. 2.8

c. 4.1 ✓

d. 5.3

$A(t) = 126e^{-\frac{1}{10}t}$. Set $84 = 126e^{-\frac{1}{10}t}$: $t = \frac{\ln(\frac{2}{3})}{-\frac{1}{10}} \approx 4.1$.

26 [Foundational] • Target: Newton's Law of Cooling

A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 9 - \frac{y}{25}$. Find the equilibrium value of y.

a. 216

b. 34

c. 225 ✓

d. 250

Setting $\frac{dy}{dt} = 0$: $9 = \frac{y}{25}$ gives $y = 9(25) = 225$ kg.

27 [Foundational] • Target: Visualizing an exponential growth solution

A population of bacteria is modeled by the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{L})$, where $P(t)$ is the population at time t in hours, k is the growth rate constant, and L is the carrying capacity. It is observed that $\frac{dP}{dt} = 200$ when $P = 100$, and the maximum growth rate occurs when $P = 500$. Find the carrying capacity L of this population.

a. **1000** ✓

b. 750

c. 1200

d. 500

The logistic differential equation is given by $\frac{dP}{dt} = kP(1 - \frac{P}{L})$. The growth rate $\frac{dP}{dt}$ is maximized when the population P is half of the carrying capacity L. We are given that the maximum growth rate occurs when $P = 500$. Therefore, $L/2 = 500$, which implies $L = 1000$. We can also use the other piece of information to verify this. When $P = 100$, $\frac{dP}{dt} = 200$. Substituting $P = 100$ and $L = 1000$ into the differential equation: $200 = k(100)(1 - \frac{100}{1000})$.

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28 [Foundational] • Target: A logistic-style differential equation

The equation $y' = ky$ models exponential growth when:

a. $k = 0$

b. $k < 0$

c. $k = 1$

d. $k > 0$ ✓

Applying Differential Equations principles, $k > 0$: growth; $k0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Additional separable equations

Consider the logistic differential equation $\frac{dP}{dt} = kP(M - P)$, where $P(t)$ is the population at time t, k is the growth rate constant, and M is the carrying capacity. If $\frac{dP}{dt} = 0.05P(1000 - P)$, what is the carrying capacity of the population?

a. **1000** ✓

b. 0.05

c. 50

d. 0.00005

The logistic differential equation is given by $\frac{dP}{dt} = kP(M - P)$. In the given equation, $\frac{dP}{dt} = 0.05P(1000 - P)$. By comparing the two forms, we can identify the carrying capacity M. We see that $k = 0.05$ and $M = 1000$. Therefore, the carrying capacity of the population is 1000.

30 [Intermediate] • Target: A mixing problem modeled by a differential equation

Coffee at 70°C is placed in a 22°C room and cools to 45°C in 5 min (Newton's law of cooling). How long (min) until it reaches 35°C?

a. 7

b. $\frac{10\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$

c. $\frac{5\ln(2)}{\ln\left(\frac{14}{9}\right)}$

d. $\frac{5\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$ ✓

$T - 22 = 48e^{-kt}$ with $k = \frac{\ln\left(\frac{48}{23}\right)}{5}$. Solve $48e^{-kt} = 13$: $t = \frac{5\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$.

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31 [Intermediate] • Target: Verifying solutions to differential equations

Newton's law of cooling: $\frac{dT}{dt} = k(T - T_s)$, $k < 0$. Steady state:

a. $T = 0$

b. $T = kT_s$

c. $T = T_s$ ✓

d. $T = T_0$

Applying Differential Equations principles, $T \to T_s$ as $t \to \infty$. Therefore, the verified correct option is $T = T_s$.

32 [Intermediate] • Target: Solving separable differential equations

Exponential decay: $\frac{dN}{dt} = -kN$, $k > 0$. Solution:

a. $N = N_0 e^{kt}$

b. $N = N_0/(1 + kt)$

c. $N = N_0 - kt$

d. $N = N_0 e^{-kt}$ ✓

Applying Differential Equations principles, $N = N_0 e^{-kt}$. Therefore, the verified correct option is $N = N_0 e^{-kt}$.

33 [Intermediate] • Target: Slope fields

Consider the differential equation $\frac{dy}{dx} = 2x$. Which of the following is a particular solution passing through the point $(1, 3)$?

a. $y = x^2 + 2$ ✓

b. $y = x^2 + 1$

c. $y = 2x + 1$

d. $y = 2x^2 - 1$

The given differential equation is $\frac{dy}{dx} = 2x$. To find the general solution, we integrate both sides with respect to x: $\int \frac{dy}{dx} dx = \int 2x dx$ $y = x^2 + C$, where C is the constant of integration. We are given that the particular solution passes through the point $(1, 3)$. Substitute these values into the general solution: $3 = (1)^2 + C$ $3 = 1 + C$ $C = 3 - 1 = 2$. Therefore, the particular solution is $y = x^2 + 2$.

34 [Intermediate] • Target: Exponential growth and decay models

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:

a. zero

b. minimum

c. **maximum** ✓

d. negative

Applying Differential Equations principles, Maximum growth at $P = K/2$. Therefore, the verified correct option is maximum.

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35 [Intermediate] • Target: Matching slope fields to differential equations

Carrying capacity K in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$

a. 0 ✓

b. K

c. ∞

d. -1

Applying Differential Equations principles, At $P = K$: $dP/dt = rK(1 - K/K) = 0$. Therefore, the verified correct option is 0.

36 [Intermediate] • Target: Newton's Law of Cooling

Initial value problem: $y' = y$, $y(0) = 2$. Solution:

a. $y = e^{2x}$

b. $y = 2e^{x}$ ✓

c. $y = 2x + 2$

d. $y = xe^{x}$

Applying Differential Equations principles, $y = Ce^{x}$; $y(0) = C = 2$. Therefore, the verified correct option is $y = 2e^{x}$.

37 [Intermediate] • Target: Visualizing an exponential growth solution

Separable ODE $\frac{dy}{dx} = xy$. Solution:

a. $y = x^2/2 + C$

b. $y = Ce^{x}$

c. $y = Ce^{x^2/2}$ ✓

d. $y = Ce^{x^2}$

$\frac{dy}{y} = x\,dx \Rightarrow \ln|y| = x^2/2 + C$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A logistic-style differential equation • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Separable ODE: $\frac{dy}{dx} = \frac{x}{y}$. Solution (implicit):

a. $x^2 + y^2 = C$

b. $y^2 = x^2$

c. $y = x + C$

d. $y^2 = x^2 + C$ ✓

$y\,dy = x\,dx \Rightarrow y^2/2 = x^2/2 + C_1$.

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39 [Challenging] • Target: Additional separable equations • Cross-Lesson: 2.01 The Definition of the Derivative

Logistic equation steady states (equilibria):

a. $P = 0$ only

b. $P = K$ only

c. $P = 0$ **and** $P = K$ ✓

d. $P = K/2$

$rP(1 - P/K) = 0 \Rightarrow P = 0$ or $P = K$.

40 [Challenging] • Target: A mixing problem modeled by a differential equation • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Consider the slope field shown below. Which of the following differential equations could represent this slope field, assuming the slopes are defined at all points except where division by zero occurs? $\frac{dy}{dx} = \frac{x}{y}$

a. $\frac{dy}{dx} = \frac{x}{y}$ ✓

b. $\frac{dy}{dx} = \frac{y}{x}$

c. $\frac{dy}{dx} = x - y$

d. $\frac{dy}{dx} = x + y$

The slope field shows that at points where $y = 0$ (the x-axis), the slopes are undefined (vertical lines). This is consistent with the differential equation $\frac{dy}{dx} = \frac{x}{y}$, as division by zero occurs when $y = 0$. Additionally, observe the slopes in different quadrants. In the first quadrant ($x > 0, y > 0$), the slopes are positive, consistent with $\frac{x}{y} > 0$. In the second quadrant ($x0$), the slopes are negative, consistent with $\frac{x}{y}0$. In the fourth quadrant ($x > 0, y < 0$), the slopes are.

41 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 3.01 The Chain Rule

Logistic growth model is $\frac{dP}{dt} =$

a. kP

b. $kP(1 - P/M)$ ✓

c. $k(M - P)$

d. P^2

Logistic: $kP(1 - P/M)$ where M is carrying capacity.

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42 [Challenging] • Target: Solving separable differential equations • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

a. $e^{-1} + 1$

b. $2e^{-1} + 1$ ✓

c. $e^{-1} - 1$

d. $2 - e^{-1}$

The given differential equation for the y-component of the velocity is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r-1)^2 = 0$. Thus, we have a repeated root $r = 1$. The general solution for $y(t)$ is $y(t) = c_1 e^t + c_2 t e^t$. Using the initial conditions $y(0) = 2$ and $y'(0) = 1$: $y(0) = c_1 e^0 + c_2(0)e^0 = c_1 = 2$. $y'(t) = 2e^t + 2te^t + c_2 e^t + c_2 t e^t = (2 + c_2)e^t + (2 + c_2)te^t$. $y'(0) = (2 + c_2)e^0 + (2 + c_2)(0)e^0 = 2 + c_2 = 1$, so $c_2 = -1$. Thus.

43 [Challenging] • Target: Slope fields • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Picard's theorem guarantees existence and uniqueness of IVP solution when f and $\partial f/\partial y$ are:

a. constant

b. polynomial

c. **continuous on a neighborhood of** (x_0, y_0) ✓

d. analytic everywhere

Applying Differential Equations principles, Picard-Lindelöf conditions. Therefore, the verified correct option is continuous on a neighborhood of (x_0, y_0).

44 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Newton's law of cooling: $T(t) = T_e + (T_0 - T_e)e^{-kt}$. If room temp $T_e = 20$, initial $T_0 = 80$, $k = 0.1$, $T(10) =$

a. $20 + 60$

b. $60e^{-1}$

c. $80e^{-1}$

d. $20 + 60e^{-1} \approx 42.1$ ✓

Applying Transcendental Functions principles, $T(10) = 20 + 60e^{-1} \approx 42.1°\text{C}$. Therefore, the verified correct option is $20 + 60e^{-1} \approx 42.1$.

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45 [Challenging] • Target: Matching slope fields to differential equations • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1e^{r_1x} + C_2e^{-r_1x}$

b. $y = C_1e^{r_1x}$

c. $y = (C_1 + C_2)e^{2r_1x}$

d. $y = (C_1 + C_2x)e^{r_1x}$ ✓

Applying Differential Equations principles, Repeated root: multiply by x. Therefore, the verified correct option is $y = (C_1 + C_2x)e^{r_1x}$.

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